

Research on Image Semantic Segmentation Algorithm Based on Fully Convolutional HED-CRF

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Abstract-Aiming at the problem that the local detail is not accurate enough in 3D scene reconstruction in complex scenes, an image semantic segmentation algorithm based on Holistically Nested Edge Detection-Conditional Random Field(HED) is proposed.First, a Fully Convolutional Networks (FCN) is trained based on Visual Geometry Group Net (VGGNet).The two-stage training method is used to realize the segmentation of the foreground and the background, and the classification image is generated to predict the semantics;Then the edge information is refined by HED algorithm. Finally, the semantic prediction results are optimized by conditional random field. The full convolution network and conditional random field can be integrated into an end-to-end system to achieve accurate segmentation of image semantics.The experimental results show that compared with the traditional machine learning method, the accuracy of this method is improved by 2.9%, and the segmentation effect is better.

Keywords- Fully Convolutional Networks (FCN); Holistically Nested Edge Detection-Conditional Random Field(HED); image semantic segmentation; machine learning

I. INTRODUCTION

Three-dimensional reconstruction is the analysis and processing of three-dimensional objects in a computer environment. Compared with one-dimensional sound data, two-dimensional image data, and three-dimensional reconstruction, the three-dimensional model is more in line with people's intuitive understanding of the real world. With the development of computer vision, people are paying more and more attention to 3D reconstruction technology. The advantages and disadvantages of 3D reconstruction are affected by the semantic segmentation of images, and image semantic segmentation is inseparable in the fields of industrial automation, unmanned vehicle driving, medical image analysis, intelligent robots, geographic information systems, etc. Segmentation has important practical significance. Prior to the advent of deep learning, image semantics were mostly based on low-order visual information of the image pixels themselves, from simple pixel-level edge-based segmentation, threshold-based segmentation, region-based segmentation, and cluster-based segmentation. , to graph-based segmentation, fusion-based segmentation, and so on.After the computing power is improved, machine learning is applied to image semantic segmentation, but the low-level semantics are obtained.These methods work well in an ideal environment.However, when it is affected by noise and light brightness, since there is no algorithm training stage, the computational complexity is not

high, and no artificial auxiliary information is provided, and the segmentation effect is not very good.In recent years, with the study of deep learning,Especially after the convolutional neural network made a major breakthrough in image classification,Semantic segmentation has also entered a new stage of development.Deep convolutional neural networks have been applied to them, and the accuracy and efficiency have been found to exceed traditional methods.Grangier et al. proposed a simple convolutional neural network (CNN) model to segment image semantics [1], which proved the feasibility of CNN in image semantic segmentation; Socher et al proposed a recurrent-based neural network. The method of the Recursive Neural Network (RNN) [2] maps the extracted image features to the semantic space through the neural network and performs autonomous learning as the input of the RNN. R.Girshick et al. propose a region-based convolution. The neural network (Region-based CNN) model [3], the target detection is to generate multiple target candidate regions, as the input image of CNN, which reduces the huge workload caused by the traditional sliding window method and accelerates the feature extraction speed. However, pooling can result in the loss of parts of the area containing important information.

The front end of this paper adopts the FCN [4] algorithm, which restores the image size through the upsampling operation to prevent the loss of important information, but the edge information of the image is rough, and it cannot guarantee that all the lost information will be returned without loss; therefore, The CRF is used as the post-processing operation. The energy function of the prediction result can be directly used to guide the training of FCN model parameters and optimize the semantic prediction results. At the same time, the traditional edge detection algorithm is combined with the deep convolutional neural network to achieve the edge. Refinement. Since the added upsampling process increases the workload, we also use the two-stage method for training, which greatly reduces the training time of the model and speeds up the convergence of the model.

II. IMAGE SEMANTIC SEGMENTATION

Image semantic segmentation is the classification of pixel points in an image and the recognition of classification results. The FCN is implemented based on the mainstream deep convolutional neural network, which is a pixel-level end-to-end semantic segmentation as a front-end coarse extraction. HED is an edge detection algorithm based on FCN used to refine edge information, and CRF is used as the output of the backend optimization front end [5]. The general framework is as follows:

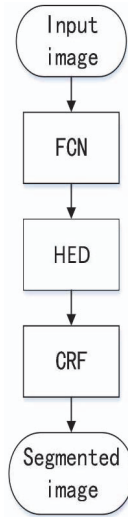


Fig.1 Image semantic segmentation framework

III. FULLY CONVOLUTIONAL HED-CRF

A. Fully Convolutional Neural Network

FCN(Full Convolutional Neural Network) mainly uses three technologies: convolution, upsampling, and skip layer .

1) Convolution

Convolution is to replace the fully connected layer of a general classification network with a convolutional layer, so that the entire network has a large amount of convolution from input to output. In this paper, VGG16 is used. All the activation units of the hidden layer are using ReLU. The convolution kernel moves one step at a time. The pooling is also one step at a time. The maximum pool is used, Conv is the convolution layer, and soft-max is the classifier. Device.The network structure table is as follows:

Tab.1 VGG16 network structure

structure	input	pooling
Conv1	224*224*3	
Conv2	224*224*64	2*2
Conv3	224*224*64	
Conv4	112*112*128	
Conv5	112*112*128	2*2
Conv6	56*56*256	
Conv7	56*56*256	
Conv8	56*56*256	2*2
Conv9	28*28*512	
Conv10	28*28*512	
Conv11	28*28*512	2*2
Conv12	14*14*512	
Conv13	14*14*512	
Conv14	14*14*512	2*2
Conv15	3*3*4096	
Conv16	3*3*4096	
Soft-max	1*1*21	

2) Upsampling

Since the image becomes smaller after pooling, partial pixel loss causes partial information incompleteness; and image semantic segmentation studies the classification result of each pixel, so it is necessary to upsample the image to restore the original image size. Upsampling is deconvolution, which turns from many-to-one to one-to-one, inverting the forward and backward propagation of the convolution. Upsampling does not guarantee that the final feature map is exactly the same size as the original image, so a crop layer is then attached to make it exactly the same size as the original image.

3) Skip layer

Since the results obtained after upsampling are rough, this structure is used to optimize the results. Image semantic segmentation is mainly classification and positioning, which divides pixels of different categories and knows their tag information. This structure is to combine the two. It combines the more accurate convolutional layer with the previous pixel points and the more accurate convolutional layer, and realizes the combination of high-resolution features and low-resolution features to achieve accurate semantic segmentation.

The specific method is: convolving and pooling the original image, and the image is reduced to 1/2 after each pooling, and after five times of pooling, the obtained feature image is 1/32 of the original image. At this time, the upsampling is performed, and the multiple is 32 to obtain the feature image of the same size of the original image. The result is called FCN-32s. The FCN-32s result is not fine enough. Based on this, fine-tuning is performed. The fourth pooling layer is added to the 2x sampling result of Conv15, and then 16 times upsampling is performed to obtain the characteristic image of the same size of the original image. Called FCN-16s. After fine tuning on the basis of FCN-16s, the third pooling layer is added to the 4 times sampling result of Conv15, and then 8 times upsampling is performed to obtain the characteristic image of the same size of the original image. The result is called FCN-8s. . Fine-tuning on the basis of FCN-8s, adding the second pooling layer and the 8 times sampling result of Conv15, and then performing 4 times upsampling, can obtain the characteristic image of the same size of the original image, and the result is called FCN-4s. Fine-tuning based on the FCN-4s, adding the first pooled layer to the 16-times sampling result of Conv15, and then performing 2 times upsampling, can obtain the feature image of the same size of the original image. The result is called FCN-2s.

The experimental results show that FCN-2s has the best effect, but the shortcomings of FCN are obvious. It takes five times to get FCN-2s, and the result is still not fine. The relationship between pixels and pixels is not considered, and the spatial consistency is neglected.

B. HED Edge Detection Algorithm

Edge detection is to perform a two-category for each pixel in the image to determine whether it is an edge. The purpose is to divide the pixels of the same type into one region. The edge detection algorithm used in this paper is global nested edge detection (Holistically-Nested Edge Detection (HED)) is based on the FCN and VGG network architecture. It is a multi-network multi-scale prediction system that outputs edges of different scales

through multiple side outputs, and then obtains the final edge output through a trained weight fusion function. The convolutional layer in the middle of the network upsamples its output to obtain an image identical to the original image, and compares the error with the reference graph. The image output from the convolutional layer in the middle is the side output. Different convolutional layers can independently learn features of different scales, and the generated errors are reversely transmitted to the corresponding convolutional layers, reducing the disappearance of the gradient. The effect of this algorithm is far superior to traditional edge detection algorithms, such as canny.

1) HED network architecture

The network of this paper is based on VGG16 and will have five layers of different scales. Outline output. It is assumed that the output of Conv2, Conv4, Conv7, Conv10, and Conv13 layers is obtained, and the output is convoluted to obtain the classification result. This paper is a two-category, that is, whether the pixel points belong to the boundary. The receptive field of the convolution layer has the same size as its corresponding side output layer. Finally, the obtained loss function is subjected to a linear weighting calculation to obtain an output result. The HED network architecture is shown below:

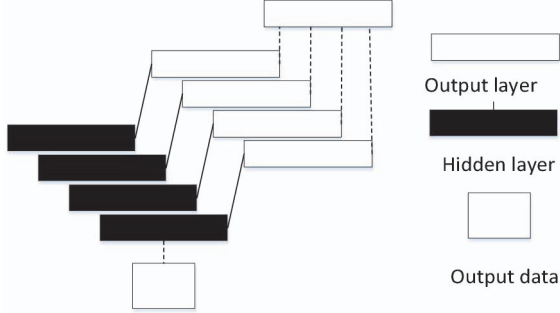


Fig.2 HED network framework

2) Network training

In the reference image, most of the pixels are not edge pixels. Therefore, it will be asymmetrical when sampling. If you use the general loss function for training, you may not be able to train it, because the network will consider that all are not part of the contour pixel, which will generate a smaller loss function, which will deviate from the actual. HED uses a class-balanced cross-entropy loss function that combines the output of five layers. This article adds a layer of weighted fusion layer, which can make use of the prediction results of side output. The final output is the aggregation of the side output layer and the weighted fusion layer. The training set and loss function are calculated as follows:

$$S = \{(X_n, Y_n)\}, X_n = \{x_j^{(n)}\}, Y_n = \{y_j^{(n)}\} \quad (1)$$

$$L_{side}(W, w) = \sum_{m=1}^M \alpha_m l_{side}^{(m)}(W, w^{(m)}) \quad (2)$$

$$L_{fuse}(W, w) = Dist(Y, \hat{Y}_{fuse}) \quad (3)$$

The final objective function is:

$$L(W, w)^* = \arg(L_{side}(W, w) + L_{fuse}(W, w)) \quad (4)$$

Where S is the training set, x is a certain pixel point, y is the corresponding label, and the value is 0 or 1; W is the parameter of the entire standard network, w is the parameter of the side output, and $w^{(m)}$ is the m th. The parameters of the convolution phase, l_{side} is the pixel-level loss function for each side of the side output, L_{side} is the linear weighted sum of the side outputs of all layers; L_{fuse} is the fusion layer loss function.

C. CRF Condition Random Field

CRF is a typical discriminant model, but at the same time has the advantage of a production model. It can avoid the problem of mark bias caused by other discriminant models. CRF models the conditional probability after multiple variables are observed. For each image, the category label of the pixel i is x_i , the corresponding observation value is y_i , each pixel point is used as a node, and the relationship between the pixel point and the pixel point as an edge constitutes a CRF. The goal of CRF is to find its conditional prediction probability $P(x_i|y_i)$.

The category label of the corresponding pixel is predicted by the observation value.

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$$P(X = x | I) = \frac{1}{Z(I)} \exp(-E(x | I)) \quad (5)$$

$$E(x) = \sum_i \psi(x_i) + \sum_{i < j} \psi(x_i, x_j) \quad (6)$$

Where I is global observation, X is the label value, $Z(I)$ is the distribution function, used for normalization, $E(x|I)$ is the energy function, $\psi(x_i)$ is a one-potential function and the output from the front-end FCN-2s. $\psi(x_i, x_j)$ is a binary potential function from the backend CRF, which describes the relationship between pixels. Similar pixels are assigned the same label, otherwise different labels are assigned. The criterion of this discriminant is related to the color and the actual relative distance, so the CRF makes the segmentation occur as much as possible at the boundary, and the segmentation of the image edge is refined.

IV. IMAGE SEGMENTATION RESULTS AND COMPARISON

A. Experimental Data and Indicators

To verify the effectiveness of the proposed algorithm, the indoor dataset uses Pascal VOC 2012, which includes 20 object classes, including 21 classes. Among them, there are about 2,913 pictures for image segmentation, including the original image, image classification and segmentation, and object segmentation. The outdoor dataset uses the Jilin University annotated data set. Commonly used evaluation indicators for image semantic segmentation are: pixel accuracy, Average Precision, Average IoU, Weighted IoU.

Pixel accuracy refers to the proportion of pixels with the correct classification; the average precision refers to the mean of the proportion of pixels with the correct classification in all labels; IoU (intersection over union) is

the ratio of the intersection of the predicted value and the true value. , that is, the overlap ratio between the target window and the markup window generated by the model.

B. Experimental Results and Analysis

The experimental results are shown in the following table. It can be seen that the algorithm in this paper works well on the Pascal VOC 2012 dataset, and the average accuracy is significantly better than FCN-8s, CRF-RNN, DeepLab3 [8].

Tab.2 Average Precision (AP %)

Algorithm	FCN-8s	CRF-RNN	Deep Lab3	ours
mean	62.2	65.2	85.7	88.6
Aero plane	76.8	80.9	96.4	97.2
Bicycle	34.2	34.0	76.6	80.1
bird	68.9	72.9	92.7	96.3
boat	49.4	52.6	77.8	80.2
Bottle	60.3	62.5	87.6	90.1
Bus	75.3	79.8	96.7	98.2
Car	74.7	76.3	90.2	95.4
Cat	77.6	79.9	95.4	97.4
Chair	21.4	23.6	47.5	57.8
Cow	62.5	67.7	93.4	95.9
Dining-Table	46.8	51.8	76.3	78.9
Dog	71.8	74.8	91.4	95.1
Horse	63.9	69.9	97.2	97.2
Motorbike	76.5	76.9	91.0	93.8
Person	73.9	76.9	92.1	93.2
Potted-Plant	45.2	49.0	71.3	78.4
Sheep	72.4	74.7	90.9	95.3
Sofa	37.4	42.7	68.9	74.2
Train	70.9	72.1	90.8	93.8
TV/Monitor	55.1	59.6	79.3	81.2

The indoor visualization of the algorithm in this paper is shown in Figure 3, and the effect is obviously better than FCN-8s and DeepLab3.

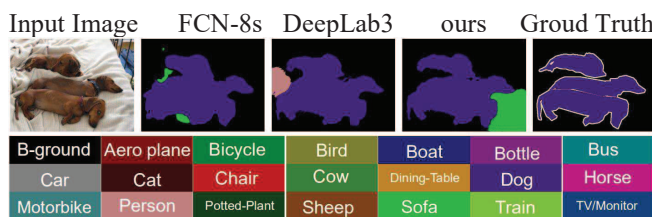


Fig.3 Visualization comparison of different algorithms

The outdoor visualization of the algorithm in this paper is shown in Figure 4, the first is the original image, the second is the segmentation obtained by the algorithm, and the third is the baseline. Although some of the details are not segmented accurately, the overall segmentation effect is good.

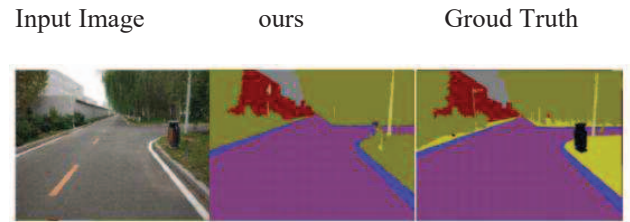


Fig.4 Outdoor visualization of our algorithm

V. CONCLUSION

Aiming at the image semantic segmentation in 3D scene reconstruction, this paper proposes an image semantic segmentation algorithm based on fully convolutional HED-CRF, and the segmentation effect is good. In the actual video image segmentation process, the obtained video images are semantically segmented by the fully convolutional neural network, and the edge detection algorithm is used to perform edge refinement. Finally, the conditional random field is used to optimize the semantic prediction results. The experimental results show that the proposed algorithm can segment the image semantics more accurately. Compared with the traditional method, the calculation is small and the accuracy is improved by 2.9%. In the next step, the edge of a specific object will be more quickly and accurately segmented in real time.

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